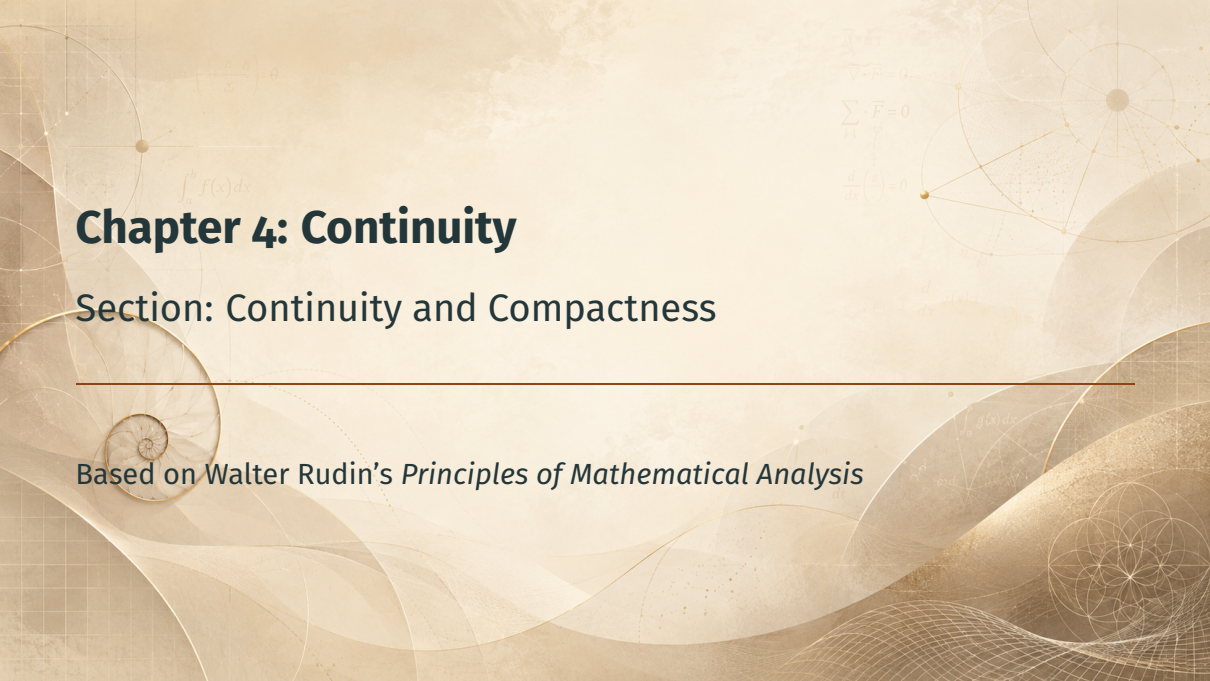


The background is a light beige color with various mathematical and geometric elements. In the top left, there is a partial view of a golden spiral. In the top right, there are faint, overlapping circles and lines, resembling a geometric construction or a network diagram. Scattered throughout are several mathematical formulas in a light, semi-transparent font. These include:  $\left( \frac{a_1}{b_1}, \frac{a_2}{b_2} \right) = \theta$  in the top left;  $\int_a^b f(x) dx$  in the middle left;  $\vec{\nabla} \cdot \vec{F} = 0$  in the top right;  $\sum_{i=1}^n \vec{F}_i = 0$  in the middle right;  $\frac{d}{dx} \left( \frac{g}{h} \right) = 0$  in the bottom right;  $\int_a^b g(x) dx$  in the bottom right; and  $\frac{d}{dt}$  in the bottom right. The main title 'Chapter 4: Continuity' is in a bold, dark blue font, and the subtitle 'Section: Continuity and Compactness' is in a regular, dark blue font. A thin horizontal line is positioned below the subtitle.

# Chapter 4: Continuity

## Section: Continuity and Compactness

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Based on Walter Rudin's *Principles of Mathematical Analysis*

# Outline

Motivation

Bounded Mappings

The Key Theorem: Compactness is Preserved

Consequences for Real Functions

Continuous Inverses

Uniform Continuity

Why Compactness Is Essential

Summary

# Motivation

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# Why Combine Continuity and Compactness?

- Continuity controls functions **locally**, point by point

$$\int_a^b f(x) dx$$



# Why Combine Continuity and Compactness?

- Continuity controls functions **locally**, point by point
- Compactness is a powerful **global** finiteness property

# Why Combine Continuity and Compactness?

- Continuity controls functions **locally**, point by point
- Compactness is a powerful **global** finiteness property
- Together they yield **classical** theorems of analysis

**Preview:** extreme value theorem, continuous inverses, and automatic uniform continuity.

$$\nabla \cdot F = 0$$

$$\frac{d}{dx}(e^x) = e^x$$

$$\int_a^b f(x) dx$$

$$e^{i\pi} + 1 = 0$$

# Bounded Mappings

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$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

# Definition 4.13: Bounded Mappings

## Definition 4.13

A mapping  $\mathbf{f}$  of a set  $E$  into  $R^k$  is **bounded** if there is a real number  $M$  such that

$$|\mathbf{f}(x)| \leq M \quad \text{for all } x \in E.$$

**Idea:** the whole image  $\mathbf{f}(E)$  fits inside a single ball of radius  $M$  centered at the origin.



$$e^{i\pi} + 1 = 0$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$y = \frac{\sin x}{x}$$

# The Key Theorem: Compactness is Preserved

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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

# Theorem 4.14: Continuous Image of a Compact Set

## Theorem 4.14

Suppose  $f$  is a continuous mapping of a **compact** metric space  $X$  into a metric space  $Y$ . Then  $f(X)$  is **compact**.

**Slogan:** *continuity preserves compactness.*

# Proof Strategy: Theorem 4.14

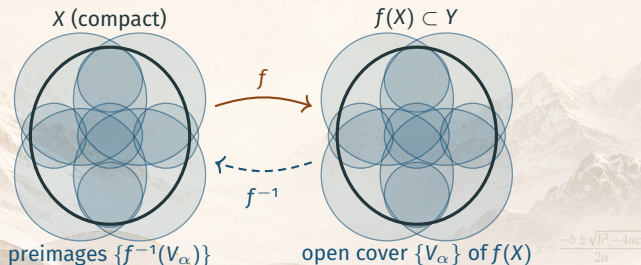
**Goal:** show every open cover of  $f(X)$  has a finite subcover.

1. Start with an open cover  $\{V_\alpha\}$  of  $f(X)$ .
2. **Pull back** each  $V_\alpha$  to get open sets  $f^{-1}(V_\alpha)$  covering  $X$ .
3. Use compactness of  $X$  to extract a **finite** subcover.
4. **Push forward** to get finitely many  $V_\alpha$  covering  $f(X)$ .

# Proof of Theorem 4.14 — Step 1: Pull Back the Cover

## Step 1: Preimages are open

Let  $\{V_\alpha\}$  be an open cover of  $f(X)$ . Since  $f$  is continuous, Theorem 4.8 shows each set  $f^{-1}(V_\alpha)$  is **open** in  $X$ .



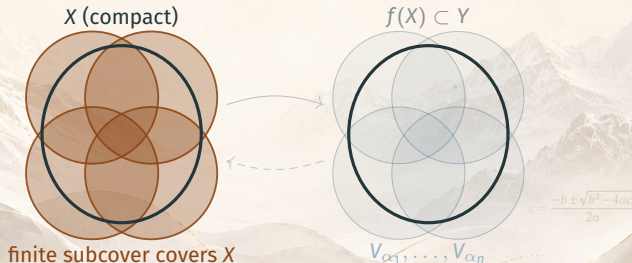


# Proof of Theorem 4.14 — Step 2: Finite Subcover in $X$

## Step 2: Compactness of $X$

The sets  $f^{-1}(V_\alpha)$  cover  $X$ . Since  $X$  is compact, finitely many indices  $\alpha_1, \dots, \alpha_n$  satisfy

$$X \subset f^{-1}(V_{\alpha_1}) \cup \dots \cup f^{-1}(V_{\alpha_n}). \quad (12)$$

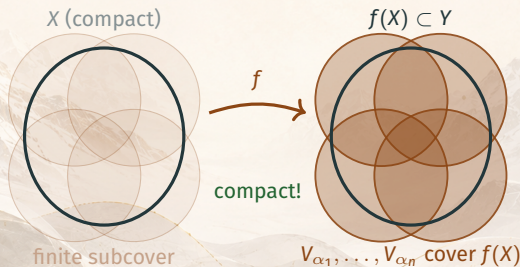


# Proof of Theorem 4.14 — Step 3: Push Forward

## Step 3: Conclusion

Since  $f(f^{-1}(E)) \subset E$  for every  $E \subset Y$ , relation (12) shows  $f(X)$  is **compact**:

$$f(X) \subset V_{\alpha_1} \cup \dots \cup V_{\alpha_n}. \quad (13)$$



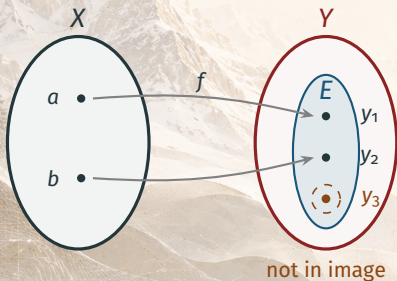
# A Useful Set-Theoretic Remark

**Remark on preimages — Equality need not hold.**

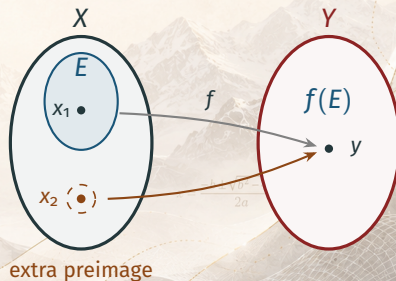
For  $E \subset Y$ :  $f(f^{-1}(E)) \subset E$ .

For  $E \subset X$ :  $f^{-1}(f(E)) \supset E$ .

$f(f^{-1}(E)) \subsetneq E$  ( $f$  not onto)



$f^{-1}(f(E)) \supsetneq E$  ( $f$  not 1-1)



The background is a complex, artistic composition. It features a landscape with rolling hills and mountains in shades of blue and grey. Overlaid on this are various mathematical and geometric elements. In the top left, there's a grid with a lightning bolt. In the top right, a complex geometric diagram with many intersecting circles and lines. In the bottom left, a large nautilus shell is shown in profile. Scattered throughout are several mathematical formulas in a light, golden-yellow color. The main title 'Consequences for Real Functions' is centered in a bold, dark blue font, with a thin horizontal line underneath it.

# Consequences for Real Functions

$$\frac{d^2 y}{dx^2} + \omega^2 y = 0$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$\nabla^2 \phi = 0$$

$$e^{im} + 1 = 0$$

$$\frac{\partial u}{\partial t} = \sigma \nabla^2 u$$

$$f(x) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$



# Theorem 4.15: Closed, Bounded, and Hence Bounded

## Theorem 4.15

If  $\mathbf{f}$  is a continuous mapping of a compact metric space  $X$  into  $R^k$ , then  $\mathbf{f}(X)$  is **closed and bounded**; thus  $\mathbf{f}$  is bounded.

**Why:** by Theorem 4.14,  $\mathbf{f}(X)$  is compact; by Theorem 2.41, compact subsets of  $R^k$  are closed and bounded.

# Theorem 4.16: The Extreme Value Theorem

## Theorem 4.16

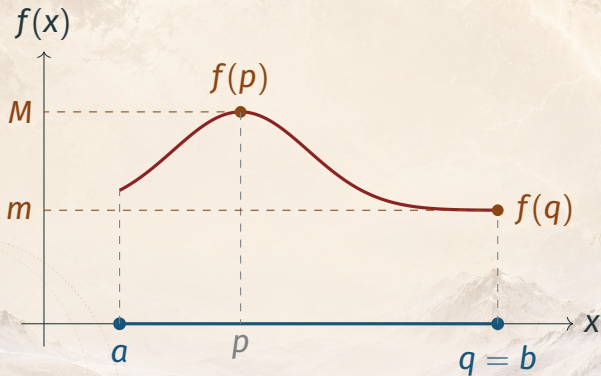
Let  $f$  be a continuous **real** function on a compact metric space  $X$ , and set

$$M = \sup_{p \in X} f(p), \quad m = \inf_{p \in X} f(p).$$

Then there exist  $p, q \in X$  with  $f(p) = M$  and  $f(q) = m$ .

Equivalently:  $f(q) \leq f(x) \leq f(p)$  for all  $x \in X$ :  $f$  **attains** its max and min.

# Visualizing the Extreme Value Theorem



# Proof of Theorem 4.16

## Proof

By Theorem 4.15,  $f(X)$  is a **closed and bounded** set of real numbers. Hence, by Theorem 2.28,  $f(X)$  contains its least upper bound and greatest lower bound:

$$M = \sup f(X) \in f(X), \quad m = \inf f(X) \in f(X).$$

So  $M$  and  $m$  are values of  $f$ . ■





$e^{i\pi} + 1 = 0$

$\phi = \frac{1+\sqrt{5}}{2}$

# Continuous Inverses

---

$\nabla^2 \psi + k^2 \psi = 0$

$\frac{d}{dx}(\sin x) = \cos x$

# Theorem 4.17: Continuity of the Inverse

## Theorem 4.17

Suppose  $f$  is a continuous **one-to-one** mapping of a compact metric space  $X$  **onto** a metric space  $Y$ . Then the inverse mapping  $f^{-1}$ , defined by  $f^{-1}(f(x)) = x$ , is continuous on  $Y$ .

$$\nabla^2 \psi + k^2 \psi = 0$$

$$\frac{d}{dx}(\sin x) = \cos x$$

**Takeaway:** on a compact domain, a continuous bijection is automatically a **homeomorphism**.

# Proof of Theorem 4.17 — Step 1: From Closed to Compact

## Step 1

By Theorem 4.8 applied to  $f^{-1}$ , it suffices to show  $f(V)$  is open in  $Y$  for every open  $V \subset X$ . Fix such a  $V$ .

The complement  $V^c$  is **closed** in  $X$ , hence **compact** (Theorem 2.35). Then  $f(V^c)$  is a compact subset of  $Y$  (Theorem 4.14).

## Proof of Theorem 4.17 — Step 2: Identify $f(V)$

### Step 2

Compact subsets of  $Y$  are **closed** (Theorem 2.34), so  $f(V^c)$  is closed. Since  $f$  is one-to-one and onto,

$$f(V) = (f(V^c))^c.$$

The complement of a closed set is open, so  $f(V)$  is **open**. ■



$$e^{i\theta} = \cos\theta + i\sin\theta$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

# Uniform Continuity

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$$\varphi = \frac{1+\sqrt{5}}{2}$$

$$\nabla^2 \phi = 0$$

$$\frac{d}{dx} \sin x = \cos x$$

# Definition 4.18: Uniform Continuity

## Definition 4.18

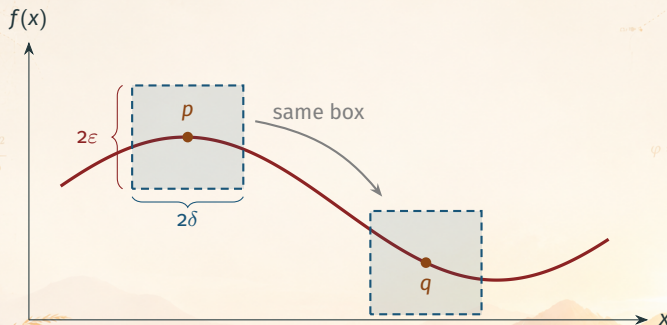
$f : X \rightarrow Y$  is **uniformly continuous** on  $X$  if for every  $\varepsilon > 0$  there exists  $\delta > 0$  such that

$$d_Y(f(p), f(q)) < \varepsilon \quad \text{whenever} \quad d_X(p, q) < \delta,$$

for **all**  $p, q \in X$ .

The same  $\delta$  works *everywhere* — it depends on  $\varepsilon$  only, not on the point.

# Visualizing Uniform Continuity



Fix  $\varepsilon$ . One  $\delta$  gives a box of width  $2\delta$ , height  $2\varepsilon$  that slides along the graph — the curve never escapes through the top or bottom.

$$\frac{d}{dx} \sin x = \cos x$$

# Continuity vs. Uniform Continuity

- **Continuity:**  $\delta$  may depend on  $\varepsilon$  **and** on the point  $p$ .

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$\varphi = \frac{1+\sqrt{5}}{2} \quad \nabla^2 \phi = 0$$

$$\frac{d}{dx} \sin x = \cos x$$



# Continuity vs. Uniform Continuity

- **Continuity:**  $\delta$  may depend on  $\varepsilon$  **and** on the point  $p$ .
- **Uniform continuity:** **one**  $\delta$  works for **all** points  $p$  at once.

Uniformly continuous  $\Rightarrow$  continuous. The converse can fail — *unless the domain is compact.*

$$\frac{d}{dx} \sin x = \cos x$$

# Theorem 4.19: Continuity on Compact $\Rightarrow$ Uniform

## Theorem 4.19

Let  $f$  be a continuous mapping of a **compact** metric space  $X$  into a metric space  $Y$ . Then  $f$  is **uniformly continuous** on  $X$ .

**Why it matters:** continuity is local, uniform continuity is global — compactness bridges the gap.

# Proof Strategy: Theorem 4.19

**Goal:** produce a single  $\delta > 0$  for a given  $\varepsilon > 0$ .

1. Continuity gives a local radius  $\phi(p)$  at each point  $p$ .
2. Cover  $X$  by the *half-size* balls  $J(p)$ .
3. Compactness: finitely many  $J(p_1), \dots, J(p_n)$  cover  $X$ .
4. Take  $\delta = \frac{1}{2} \min\{\phi(p_1), \dots, \phi(p_n)\} > 0$ .
5. Triangle inequality closes the argument.

$$\frac{d}{dx} \sin x = \cos x$$

# Proof of Theorem 4.19 — Step 1: Local Radii

## Step 1: Use continuity at each point

Let  $\varepsilon > 0$ . By continuity, to each  $p \in X$  assign  $\phi(p) > 0$  with

$$d_X(p, q) < \phi(p) \implies d_Y(f(p), f(q)) < \frac{\varepsilon}{2}. \quad (16)$$

$$\frac{d}{dx} \sin x = \cos x$$



# Proof of Theorem 4.19 — Step 2: Half-Size Cover

## Step 2: An open cover by half-balls

Let  $J(p) = \{q \in X : d_X(p, q) < \frac{1}{2}\phi(p)\}$ .

Since  $p \in J(p)$ , the collection  $\{J(p)\}$  is an **open cover** of  $X$ .

$$\frac{d}{dx} \sin x = \cos x$$

## Proof of Theorem 4.19 — Step 3: Finite Subcover, Choose $\delta$

### Step 3: Compactness gives $\delta$

By compactness, finitely many points  $p_1, \dots, p_n$  give

$$X \subset J(p_1) \cup \dots \cup J(p_n). \quad (18)$$

Put  $\delta = \frac{1}{2} \min\{\phi(p_1), \dots, \phi(p_n)\} > 0$ .

**Crucial:** the min of *finitely* many positive numbers is positive; the inf of infinitely many could be 0.

$$\frac{d}{dx} \sin x = \cos x$$

# Proof of Theorem 4.19 — Step 4: Triangle Inequality

## Step 4: One $\delta$ works for all

Take  $p, q \in X$  with  $d_X(p, q) < \delta$ . Some  $m$  has  $p \in J(p_m)$ , so  $d_X(p, p_m) < \frac{1}{2}\phi(p_m)$ , and

$$d_X(q, p_m) \leq d_X(p, q) + d_X(p, p_m) < \delta + \frac{1}{2}\phi(p_m) \leq \phi(p_m).$$

Both  $p, q$  lie within  $\phi(p_m)$  of  $p_m$ , so by (16):

$$d_Y(f(p), f(q)) \leq d_Y(f(p), f(p_m)) + d_Y(f(q), f(p_m)) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \quad \blacksquare$$



# Why Compactness Is Essential

$$\nabla \cdot \vec{F} = 0$$

$$\int_C \vec{F} \cdot d\vec{r} = 0$$

$$\frac{d}{dx}$$

$$\phi = \frac{1 + \sqrt{5}}{2}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$



# Theorem 4.20: Failures on Noncompact Sets

## Theorem 4.20

Let  $E \subset \mathbb{R}^1$  be **noncompact**. Then:

- (a) some continuous function on  $E$  is **unbounded**;
- (b) some continuous *bounded* function on  $E$  has **no maximum**;
- (c) if  $E$  is also bounded, some continuous function on  $E$  is **not uniformly continuous**.

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

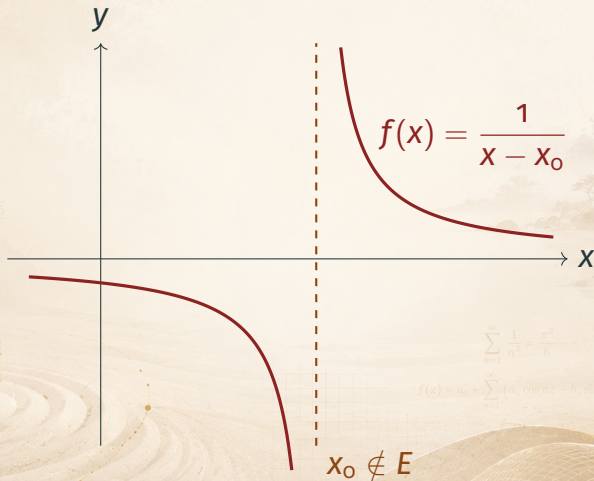
## Proof of 4.20 — Bounded Noncompact $E$

$E$  bounded but not compact  $\Rightarrow$  has a limit point  $x_0 \notin E$ .

$$f(x) = \frac{1}{x - x_0} \quad \text{cont., **unbounded**, not uniformly cont.} \quad (21)$$

$$g(x) = \frac{1}{1 + (x - x_0)^2} \quad \text{cont., bounded, } \sup g = 1 \text{ **not attained**} \quad (22)$$

# Visualizing the Blow-Up at $x_0$



## Proof of 4.20 — Unbounded $E$ and Part (c)

**If  $E$  is unbounded:**

$f(x) = x$  gives (a),  $h(x) = \frac{x^2}{1+x^2}$  gives (b) ( $\sup h = 1$  not attained).

**Part (c) needs boundedness:** on  $E = \mathbb{Z}$ , take  $\delta < 1$ : *every* function is uniformly continuous.

## Example 4.21: A Continuous Bijection with Bad Inverse

### Example 4.21

Let  $X = [0, 2\pi)$  and let  $\mathbf{f}$  map  $X$  onto the unit circle  $Y$ :

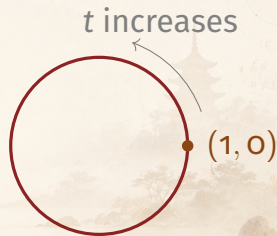
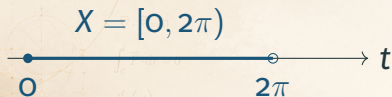
$$\mathbf{f}(t) = (\cos t, \sin t), \quad 0 \leq t < 2\pi.$$

$\mathbf{f}$  is a continuous 1-1 map onto  $Y$ , but  $\mathbf{f}^{-1}$  is not continuous at  $(1, 0)$ .

**Reason:**  $X = [0, 2\pi)$  is not compact.



# Visualizing Example 4.21: The Seam



Points just *below*  $2\pi$  map near  $(1, 0)$ , but their preimages are far from  $0$  — the inverse **tears** the circle.


$$\varphi = \frac{1+\sqrt{5}}{2}$$

$$\frac{d}{dx} \phi(x) = 0$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$\nabla \cdot \vec{F} = 0$$

$$\int_a^b f(x) dx$$

# Summary

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$$\frac{a}{b} = \frac{b}{a+b} = \varphi$$

$$F_n = F_{n-1} + F_{n-2}$$

$$\frac{d}{dt} E = 0$$

# Summary

## Key takeaways:

- **Theorem 4.14:** continuity preserves compactness —  $f(X)$  is compact.

# Summary

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- **Theorems 4.17 & 4.19:** continuous bijections invert continuously; continuity becomes *uniform*.



# Summary

## Key takeaways:

- **Theorem 4.14:** continuity preserves compactness —  $f(X)$  is compact.
- **Theorem 4.16:** a continuous real function on a compact set *attains* its max and min.
- **Theorems 4.17 & 4.19:** continuous bijections invert continuously; continuity becomes *uniform*.
- **Theorems 4.20–4.21:** drop compactness and every result can fail.

**Coming up next:** *Continuity and Connectedness.*